

EcoVault Face Unlock for Field Research Data

I am designing a face unlock system for an app called EcoVault. Environmental researchers use this app on rugged laptops and phones to store sensitive data like endangered species locations, field notes, and water quality surveys. The goal is to use a standard RGB camera to verify the user right after they enter a PIN, so the app adds a second layer of protection without slowing people down too much.

Persona & Environment

The main users are field biologists and ecologists working in places like wetlands, forests, park sites, and temporary field camps. They use rugged laptops and normal smartphones that may only have a standard RGB camera. The environment is inconsistent and hard to control. Sometimes they are in bright sunlight, sometimes in shade, and sometimes in dim tents or vehicles. They may also be wearing hats, sunglasses, masks, or rain gear. Because they often have no cell service, the full system has to run offline on the device.

Risk Level & Threats

The risk level is moderate to high. The data stored in EcoVault can be sensitive enough that unauthorized access could expose endangered species locations or unpublished research. The main threats are realistic ones like a coworker, guide, or thief picking up an unattended device, knowing or guessing the PIN, and attempting to unlock the app with their own face or a simple photo of the enrolled user. I am focusing on these common unauthorized access attempts rather than advanced attacks like custom masks.

Privacy & Safety Constraints

Privacy is a major part of this use case. All processing should happen locally on the device. No images or biometric data should be sent to a server. The system should store encrypted face embeddings instead of raw photos, present a clear consent screen during enrollment, and let the user fully delete their biometric profile and any logs at any time. The camera should only activate during an intentional unlock attempt and never run in the background.

Acceptance Targets

The main metric for this project is TPR at a fixed FAR. I care more about that than plain accuracy because this is an authentication task with class imbalance, not a balanced classification problem. Since EcoVault uses face verification after a PIN, I am treating it as a second-factor unlock. Under normal conditions, my target is TPR of at least 90% at FAR of at most 2%. Under harder conditions like darker lighting or partial face covering, my target is TPR of at least 80% at FAR of at most 5%. I also want EER of 8% or lower and AUC of at least 0.98.

For usability, I want the system to run at 100 ms or less on median inference, with 300 ms as a hard upper limit. That keeps the face check fast enough that it still feels worth using after a PIN. For deployment size, I want the model to stay under 20 MB if possible, with 25 MB as a hard cap, since field devices need space for offline maps, notes, photos, and survey data.